

CL677 : Modelling Stochastic and Turbulent Transport

(Spring 2025-26)

Assignment 2

(total points: 55+5 = 60)

Important Note: Students are expected to actively collaborate with your group members and solve this assignment. Each group has to submit one PDF file with your detailed answers to this assignment. There will be an online viva session in which each member will be assessed individually; every member is expected to have an indepth understanding of the submitted work and to be able to explain/justify any step in the calculation or code. The individual marks awarded during the viva will be used to scale the marks awarded to the group-submission. Copying across groups is prohibited; if detected, both groups involved in copying (the lender and the copier) will be penalized with zero marks. So do not share codes or answers across groups.

Question A: Consider the following Fokker-Plank equation for the p.d.f. $p(x, t)$ of a Brownian particle located at position x at time t , with a constant drift velocity A and diffusion coefficient D .

$$\frac{\partial p}{\partial t} = -A \frac{\partial p}{\partial x} + \frac{D}{2} \frac{\partial^2 p}{\partial x^2}.$$

The spatial domain extends from $-L/2$ to $+L/2$ with no-flux (also known as repelling) boundaries at both ends:

$$-Ap + \frac{D}{2} \frac{\partial p}{\partial x} = 0., \quad \text{at } x = -L/2, +L/2.$$

Develop a code to simulate this PDE. Use the finite difference method to discretize the spatial derivatives (choose central second-order difference formulas for both derivatives). This will yield a set of N ODEs if you choose a spatial grid with N equally spaced point. Integrate this system of coupled equations in time using any standard explicit integration scheme of second order or higher, such as the Runge-Kutta method; you can use the in-built ODE solvers in Matlab (e.g., ode45) or Python (solveivp) or Mathematica (ndsolve). This technique of solving parabolic PDEs is called the method of lines. There are several tutorials online, as well as in books on numerical methods, which demonstrate how to implement this numerical scheme for the diffusion equation on a 1D no-flux domain. Care should be taken when incorporating the boundary conditions; I suggest using ghost nodes so that the same discretized equation can be applied to all points of the domain.

Take $D = 1$ and $L = 10$. Use $N = 100$ and increase if needed for accuracy. The time step Δt will be auto-set by the integration scheme if you use an in-built routine. If you have to set it, then recall that the smallest time-scale of the system is the time for diffusion to occur at the grid-scale, i.e., $(\Delta x)^2/D$, where $\Delta x = L/(N - 1)$. The time step should be comparable to or smaller than this time, roughly speaking. In addition, if $A \neq 0$, then the time step should also satisfy the CFL criteria, $\Delta t < \Delta x/A$.

Set the initial distribution $p(0, t)$ to be a square pulse of width ℓ centered at $x = 0$, in order to approximate a delta-function initial condition. So, set $p = 0$ at all grid points, except those between $[-\ell/2, +\ell/2]$ which should be assigned a constant value such that the area of the pulse equals unity (required by the normalization condition for a p.d.f.).

In the questions that follow, you will have to simulate the FP equation with different parameter values and initial conditions. Evolve the solution until the p.d.f. reaches a steady-state (this may need a final time of $T = 10$ or more).

1. Simulate the FP equation with $A = 0$. Take $\ell = 1$.
Plot the solution, i.e. the profile of $p(x, t)$, as a function of x at different times. Overlap the curves onto a single panel so as to apprehend the evolution of the p.d.f. Plot one panel with profiles at intervals of 0.2 to see the early-time behaviour, and another panel with intervals of 1.0 to see the long-time behaviour. (10)
2. Fit the profiles to a Gaussian by calculating the variance of the profiles and then plotting a Gaussian curve with the same variance. Overlap the Gaussian fits on top of the numerical solution. Obviously the solution at very early times will not match a Gaussian because of our pulse initial condition. Does the solution begin to look like a Gaussian at later times? How soon does this happen? After this, does the solution always remain Gaussian or does it depart at long-times? Why? (5)
3. Comment on the final steady-state attained by the p.d.f. What does this mean physically? Could you predict this without solving the equations? (3)
4. Simulate the FP with $A = 1$. Run two simulations with different initial conditions corresponding to $\ell = 1$, and $\ell = L$ (uniform initial distribution). Overlap the profiles from the two simulations to compare their evolution. Do you obtain the same final steady state for both initial conditions? How does the final state differ from the case with zero drift? What is the difference between this final state and that corresponding to a case wherein $A = 1$ but $D = 0$? (10)

Question B: Consider a Brownian particle that is acted on by an external force which pulls it towards the origin. If the distance from the origin remains small, then the force-field can be linearized, which in turn will result in a deterministic drift velocity of $A = -ax$ (where a is the ratio of the force amplitude to the drag coefficient). This is equivalent to saying that the particle is in a harmonic potential well (with a parabolic profile). A simple physical situation would be a particle in a bowl, but typically particles that are small enough to be diffusive are not strongly affected by gravity. In experiments, an effective external force-field may be generated using a beam of light that can trap tiny Brownian particles. By studying the motion of the trapped particle one can infer the properties of the particle or the solvent. The FP equation for this situation is

$$\frac{\partial p}{\partial t} = \frac{\partial}{\partial x}(pax) + \frac{D}{2} \frac{\partial^2 p}{\partial x^2}.$$

Let $D = 1$ and use the same numerical setting and scheme as in question A. Start with a broad pulse initial condition with $\ell = 2$.

1. Simulate this equation for $a = 0.2, 0.5, 1$ and plot the corresponding p.d.f.s. For each case, visualize the evolution of p with time. How does the magnitude of the drag coefficient ($\propto a^{-1}$) affect the results? How does the presence of a force fundamentally change the long-term behaviour? (10)

Question C: Consider the FP equation of question A but this time with a repelling boundary on the left and an absorbing boundary on the right ($p(L/2, t) = 0$). Assume a pulse initial condition with $\ell = 1$.

1. Perform simulations for $A = 0, 0.2$ and 1 . The drift will drive particles towards the right where they will be absorbed. So, over time, the total probability of finding a particle anywhere within the domain will decrease. Calculate and plot this decrease for all three cases. (10)

2. Does the probability vs time decay curve change fundamentally with and without a drift velocity? You may want to compare the different cases using linear-linear plots, log-linear, and log-log plots to see if there is a power-law or exponential or linear decay. (5)
3. How does the decay vs time curve differ from the case where there is no diffusion, i.e., explain the effect of diffusion on the likelihood of the particle escaping. (2)

Writing and plotting: 5 additional marks are allotted for evaluating the comprehensiveness of the work and the quality of the report. Well-reasoned discussions and explanations of results, and clear and elegant plots will be rewarded. (5)